

# Long Yin (Nate) Chu

+852 6193 4952 | +44 7763 246314 | natechu65@gmail.com

LinkedIn: <https://www.linkedin.com/in/long-yin-chu-92577a226/>

Previous Work: <https://lycoda.github.io>

Hong Kong | London, United Kingdom

---

## EDUCATION

### Master of Science , UCL

Medical Robotics and Artificial Intelligence

Sept 2025 - Sept 2026

### Bachelor of Engineering, City St. George's University

Biomedical Engineering

Sept 2022 - Jun 2025

---

## KEY COMPETENCIES

**Programming & Analytics** - Python (NumPy , SciPy , PyTorch) , Git/GitHub

**Languages** - English (Native) , Mandarin (Fluent) and Cantonese (Fluent)

---

## PROJECTS

### UCL Dissertation

#### Machine Learning Classification of fNIRS Data

Dec 2025 - Present

- Developed a biomedical machine learning pipeline to classify between neurodivergent and neurotypical participants groups using fNIRS data from complex walking-task experiments.
- Using SPM beta-derived ROI, task-context, and contrast-based feature sets to represent haemodynamic responses across experimental conditions.
- Evaluated classifier performance using cross-validation, with attention to class imbalance, and participant-level generalisation.

### Robotic Pick-and-Place Perception Pipeline

#### Robotic Sensing, Manipulation and Interaction UCL Module 2026

Jan 2026 - May 2026

- Developed a ROS2-based robotic manipulation pipeline for simulated pick-and-place tasks using Panda robot in Gazebo.
- Designed task-specific service nodes to solve object matching, shape recognition and planning/execution tasks under variable object orientation and size.

### Surgical Data Science and Disease Classification Pipeline

#### Surgical Data Science UCL Module 2026

Jan 2026 - May 2026

- Developed an end-to-end machine learning workflow for biomedical disease classification, including data preprocessing, feature binarisation, dimensionality reduction, and model evaluation.
  - Applied supervised and unsupervised learning methods, including classification models.
  - Evaluated model performance using confusion matrices, accuracy, precision, recall, and F1-score, with attention to class imbalance and interpretability of diagnostic outputs.
- 

## PROFESSIONAL EXPERIENCE

### Koln 3D

#### Summer Intern

Jun 2024 - Aug 2024

- Developed a REDCap-based automated clinical follow-up system in collaboration with clinical teams, using orthopaedic outcome scoring methods to quantitatively assess implant performance and patient feedback.
- Built an automated production Gantt chart system using Excel VBA and Power Automate, enabling end-to-end visibility of R&D-to-manufacturing timelines.

### Golden Cedar Ltd

#### Summer Intern

Jun 2023 - Aug 2023

- Supported full order-to-delivery supply chain operations, coordinating product selection, quality control, invoicing, and logistics for high-volume B2B textile orders.
- Assisted in fulfilling major supply contracts for international brands including Carhartt WIP, Chef Works Inc.

### Audrey

#### Marketing Intern

Jun 2022 - Aug 2022

- Planned and executed Meta advertising campaign, leveraging AI creative tools to design visual assets for time critical inventory clearance and store closure campaign.
- Developed and implemented product bundling and strategic re-pricing initiatives offering attractiveness and driving a 200% uplift in sales whilst efficiently clearing excess inventory.
- Supported operational transition from physical retail to e-commerce, contributing to both commercial growth and backend fulfillment efficiency.